Supplementary material: Hydrostatic Mass Bias and Emulated Cluster Profile Evolution Between $z = 0$ and $z = 0.4987$

HEP-Genesis-Agent session record
Project: ramp-combo-cluster-profiles

2026-10-20

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `d937dab8`
- Started: 2026-10-20 14:31:28; report generated: 2026-10-20 14:40:51
- Project: ramp-combo-cluster-profiles (`/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles`)
- Compute execution site(s) selected in the session: local
- 1 user turn(s), 91 tool call(s) (0 returned errors), 4 figure(s), 18 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
report	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`

Server	Transport	Location
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`serve-sbbn.sh list_sterile_nu_skills`	`{}`	`{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...`
2	`serve-hydro.sh list_observables`	`{}`	`{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...`
3	`hep-research-aid list_agent_files`	`{}`	`[]`
4	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
5	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
6	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...`
7	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "hydrostatic mass bias galaxy clusters simulations", "max_results": 8}	[2409.07849v2] Hydrostatic mass bias for galaxy groups and clusters in the FLAMINGO simulations Abstract: The masses of galaxy clusters are commonly measured from X-ray observations under the assumpti...
9	local search_arxiv_ abstracts	{"query": "emulator galaxy cluster gas density temperature profiles hydrodynamical simulations", "max_results": 8}	[2101.07194v1] Magnetic properties of wurtzite (Ga,Mn)As Abstract: Here we report on detailed studies of the magnetic properties of the wurtzite (Ga,Mn)As cylindrical shells. Ga$_ {0.94}$Mn$_{0.06}$As ...
10	local search_arxiv_ abstracts	{"query": "redshift evolution hydrostatic bias non-thermal pressure support clusters", "max_results": 6}	[1507.04338v2] Analytical model for non-thermal pressure in galaxy clusters - III. Removing the hydrostatic mass bias Abstract: Non-thermal pressure in galaxy clusters leads to underestimation of the ...
11	local search_arxiv_ abstracts	{"max_results": 6, "query": "emulator thermodynamic profiles gas density temperature CAMELS feedback redshift"}	[2101.07194v1] Magnetic properties of wurtzite (Ga,Mn)As Abstract: Here we report on detailed studies of the magnetic properties of the wurtzite (Ga,Mn)As cylindrical shells. Ga$_ {0.94}$Mn$_{0.06}$As ...
12	local search_arxiv_ abstracts	{"max_results": 6, "query": "hydrostatic mass bias Planck weak lensing calibration 1-b"}	[0604069v1] The Scientific Programme of Planck Abstract: For 40 years, the cosmic microwave background (CMB) has been the most important source of information about the geometry and contents of the ...

#	Tool	Arguments	Outcome
13	local search_arxiv_ abstracts	{"max_results": 6, "query": "self-similar evolution intracluster medium density temperature profiles redshift"}	[1811.01955v2] Enrichment of the hot intracluster medium: numerical simulations Abstract: The distribution of chemical elements in the hot intracluster medium (ICM) retains valuable information about ...
14	local web_search	{"query": "emulator CGD CTP gas density temperature profiles galaxy clusters hydrodynamics"}	We introduce a universal fitting formula for the slope and amplitude of the gas density profile as a function of mass and redshift. The best-fit functions are suitable for all feedback variants consid...
15	local search_arxiv_ abstracts	{"max_results": 6, "query": "X-COP hydrostatic masses non-thermal pressure Ettori Eckert"}	[1805.00035v3] Hydrostatic mass profiles in X-COP galaxy clusters Abstract: We present the reconstruction of hydrostatic mass profiles in 13 X-ray luminous galaxy clusters that have been mapped in the...
16	local search_arxiv_ abstracts	{"max_results": 6, "query": "machine learning emulator intracluster medium radial profiles baryonic feedback parameters"}	[2306.04338v1] Changing Data Sources in the Age of Machine Learning for Official Statistics Abstract: Data science has become increasingly essential for the production of official statistics, as it en...
17	local search_arxiv_ abstracts	{"max_results": 5, "query": "universal fitting formula slope amplitude gas density profile mass redshift feedback variants"}	[2409.05815v3] The impact of feedback on the evolution of gas density profiles from galaxies to clusters: a universal fitting formula from the Simba suite of simulations Abstract: The radial distribut...

#	Tool	Arguments	Outcome
18	local search_arxiv_ abstracts	{"max_results": 5, "query": "CAMELS emulator thermal Sunyaev-Zeldovich profiles halos gas density pressure"}	[2101.07194v1] Magnetic properties of wurtzite (Ga,Mn)As Abstract: Here we report on detailed studies of the magnetic properties of the wurtzite (Ga,Mn)As cylindrical shells. Ga$_ {0.94}$Mn$_{0.06}$As ...
19	local get_arxiv_ metadata	{"arxiv_id": "2409.07849"}	arXiv ID: 2409.07849 Title: Hydrostatic mass bias for galaxy groups and clusters in the FLAMINGO simulations Authors: Joey Braspenning, Joop Schaye, Matthieu Schaller, Roi Kugel, Scott T. Kay Year: 20...
20	serve-hydro.sh list_observables	{}	{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...
21	serve-hydro.sh describe_ parameters	{"stat_name": "CGD"}	{ "status": "success", "message": "7-parameter design space for CGD: kappa_w, e_w, M_seed, v_ kin, epsilon_kin, omega_ m, sigma_8.", "files": [], "metadata": { "parameters": { "kappa_ w...
22	serve-hydro.sh list_observables	{}	{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...

#	Tool	Arguments	Outcome
23	serve-hydro.sh describe_parameters	{"stat_name": "CGD"}	{ "status": "success", "message": "7-parameter design space for CGD: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { "kappa_w...
24	serve-hydro.sh predict_observable	{"z": 0, "stat_name": "CGD", "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "omega_m": 0.1375, "sigma_8": 0.8, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluste...	{ "status": "success", "message": "Predicted CGD at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.0...
25	local download_arxiv_full_text	{"arxiv_id": "2409.07849", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/2409.07849.txt
26	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.0000_prediction.csv"}	# label: CGD z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CGD # redshift: 0.0 # x_label: r/R_{500c} # y_label: $\rho_{\ma...
27	local download_arxiv_full_text	{"arxiv_id": "1805.00034", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/1805.00034.txt
28	local download_arxiv_full_text	{"arxiv_id": "1404.4636", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/1404.4636.txt

#	Tool	Arguments	Outcome
29	serve-hydro.sh predict_observable	{"stat_name": "CGD", "z": 0, "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluste...	{ "status": "success", "message": "Predicted CGD at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.0...
30	serve-hydro.sh predict_observable	{"stat_name": "CGD", "z": 0.4987, "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-c...	{ "status": "success", "message": "Predicted CGD at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.4...
31	serve-hydro.sh predict_observable	{"stat_name": "CTP", "z": 0, "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluste...	{ "status": "success", "message": "Predicted CTP at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.0...
32	serve-hydro.sh predict_observable	{"stat_name": "CTP", "z": 0.4987, "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-c...	{ "status": "success", "message": "Predicted CTP at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.4...
33	board post_dead_end	{"what": "Predicting CGD/CTP at group-scale (~1e13.5 Msun) and cluster-scale (~1e14.5-15 Msun) masses to test the mass dependence of the slope-driven HSE bias.", "why": "The CGD/CTP/CPP emulators have no mass input. The ...	Recorded dead end x-01.

#	Tool	Arguments	Outcome
34	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.0000_prediction.csv"}	# label: CGD z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CGD # redshift: 0.0 # x_label: r/R_{500c} # y_label: $\rho_{\ma...
35	board post_question	{"text": "What halo-mass selection (M500c range or stack) defines the HACC CGD/CTP emulator profiles? The emulator metadata does not document it, and without it a group-versus-cluster comparison with FLAMINGO (2409.07849...	Recorded question q-06.
36	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.4987_prediction.csv"}	# label: CGD z=0.4987 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CGD # redshift: 0.4987 # x_label: r/R_{500c} # y_label: $\rho_{...
37	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.0000_prediction.csv"}	# label: CTP z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CTP # redshift: 0.0 # x_label: r/R_{500c} # y_label: $T / T_{\m...
38	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.4987_prediction.csv"}	# label: CTP z=0.4987 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CTP # redshift: 0.4987 # x_label: r/R_{500c} # y_label: $T / T_...

#	Tool	Arguments	Outcome
39	local download_arxiv_full_text	{"arxiv_id": "2409.05815", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/2409.05815.txt
40	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/2409.07849.txt", "length": 5000}	[chars 0-5000 of 95796] (more available, next start=5000) Title: Hydrostatic mass bias for galaxy groups and clusters in the FLAMINGO simulations Authors: Joey Braspenning, Joop Schaye, Matthieu Scha...
41	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/1805.00034.txt", "length": 4000}	[chars 0-4000 of 70252] (more available, next start=4000) Title: Non-thermal pressure support in X-COP galaxy clusters Authors: D. Eckert, V. Ghirardini, S. Ettori, E. Rasia, V. Biffi, E. Pointecoute...
42	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/1404.4636.txt", "length": 4000}	[chars 0-4000 of 45574] (more available, next start=4000) Title: Hydrodynamic Simulation of Non-thermal Pressure Profiles of Galaxy Clusters Authors: Kaylea Nelson, Erwin T. Lau, Daisuke Nagai Year: ...
43	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/2409.05815.txt", "length": 4000}	[chars 0-4000 of 108586] (more available, next start=4000) Title: The impact of feedback on the evolution of gas density profiles from galaxies to clusters: a universal fitting formula from the Simba...

#	Tool	Arguments	Outcome
44	board post_finding	{"claim": "FLAMINGO (Braspenning et al. 2024, arXiv:2409.07849): fiducial median b_HSE = (M_HSE,500c - M500c)/M500c goes from about -0.1 (groups) to about -0.2 (clusters). Scatter at fixed true mass is smallest, sigma(b)...	Recorded finding f-001.
45	board post_finding	{"claim": "X-COP (Eckert et al. 2018, arXiv: 1805.00034), 13 nearby clusters with XMM+Planck: median non-thermal pressure fraction ~6% at R500 and ~10% at R200. This is lower than hydrodynamical simulations predict. With ...	Recorded finding f-002.
46	board post_finding	{"claim": "Nelson, Lau & Nagai 2014 (arXiv: 1404.4636): the non-thermal pressure fraction profile is universal across redshift when radii are scaled to the mean matter density (R200m), not to the critical density. It depe...	Recorded finding f-003.
47	board post_finding	{"claim": "Simba (Sorini et al. 2024, arXiv:2409.05815), six feedback variants, 1e11 < M200c < 1e14 Msun, 0 < z < 4: gas-density profiles follow a power law. At fixed halo mass, slope and amplitude depend only weakly on ...	Recorded finding f-004.
48	local cite	{"arxiv_id": "2409.07849", "note": "[\"At a fixed true mass, the hydrostatic masses increase (decrease) with redshift on group (cluster) scales\", \"The bias is also sensitive to the strength of stellar and AGN feedback....	Cited: Hydrostatic mass bias for galaxy groups and clusters in the FLAMINGO simulations (\cite{arxiv240907849})

#	Tool	Arguments	Outcome
49	local cite	{"arxiv_id": "1805.00034", "note": "[\"We find that hydrostatic masses require little correction\", \"with some evidence for a mass-dependent bias\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-c...	Cited: Non-thermal pressure support in X-COP galaxy clusters (\cite{arxiv180500034})
50	local cite	{"arxiv_id": "1404.4636", "note": "[\"is universal across redshift when galaxy cluster radii are\", \"strongly dependent on the mass accretion rate of the galaxy cluster\"]", "bib_path": "/Users/nramachandra/hep-genesis-...	Cited: Hydrodynamic Simulation of Non-thermal Pressure Profiles of Galaxy Clusters (\cite{arxiv14044636})
51	board post_finding	{"claim": "HACC emulator at design-midpoint parameters (kappa_w=3, e_w=0.6, M_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8). Profiles are only available for z<=0.4987, so the pair used is z=0 and z=0.4...	Recorded finding f-005.
52	local cite	{"arxiv_id": "2409.05815", "note": "[\"generally weakly dependent on redshift\", \"Once AGN-driven jets are activated in the simulation, the gas density\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/...	Cited: The impact of feedback on the evolution of gas density profiles from galaxies to clusters: a universal fitting formula from the Simba suite of simulations (\cite{arxiv240905815})
53	local get_arxiv_metadata	{"arxiv_id": "2511.10730"}	arXiv ID: 2511.10730 Title: The signature of major mergers on the hydrostatic mass bias of galaxy clusters Authors: Isac Barranco-Llorca, David Vallés-Pérez, Susana Planelles, Vicent Quilis Year: 2025...

#	Tool	Arguments	Outcome
54	serve-hydro.sh plot_observable_comparison	{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128", "prediction_files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/26...	{ "status": "success", "message": "Plotted 2 CGD predictions.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_comparison.png... [Fig. 9]
55	serve-hydro.sh plot_observable_comparison	{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128", "prediction_files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/26...	{ "status": "success", "message": "Plotted 2 CTP predictions.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_comparison.png... [Fig. 10]
56	board post_finding	{"claim": "Barranco-Llorca et al. 2025 (arXiv: 2511.10730), AMR cluster mergers at 0 <= z <= 1.5: at z=0 the HSE bias correlates only mildly with dynamical state. During a major merger the bias dips sharply negative aroun...	Recorded finding f-006.
57	board post_question	{"text": "Still to extract: the numerical Nelson et al. 2014 f_nth(r/R200m) fit parameters (1404.4636 Sec. results), the FLAMINGO per-mass-bin b(z) values and figure number (2409.07849), the X-COP universal profile param...	Recorded question q-07.

#	Tool	Arguments	Outcome
58	board post_finding	{"claim": "Comparison overlays of the HACC emulator profiles at z= 0 and z=0.4987 (midpoint parameters). In CGD (rho_ gas/rho_crit(z)) the two redshifts agree within about 3% over 0.2-0.9 R500c. At z=0.4987 the density is ...	Recorded finding f-007.
59	hep-research-aid list_files	{"directory": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128"}	Files in /Users/ nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128: - 1404.4636.pdf - 1404.4636.txt - 1805.00034.pdf - 1805.00034.txt - 2409.05815.pdf ...
60	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128/ 1404.4636.txt", "start": 12000, "length": 12000}	[chars 12000-24000 of 45574] (more available, next start=24000) note that the choice of the redshift z = 0.5 is arbitrary, and in fact Γ is not the only (or necessarily the best) way of characterizi...
61	serve-hydro.sh describe_ parameters	{"stat_name": "CGD"}	{ "status": "success", "message": "7-parameter design space for CGD: kappa_w, e_w, M_seed, v_ kin, epsilon_kin, omega_ m, sigma_8.", "files": [], "metadata": { "parameters": { "kappa_ w...
62	serve-hydro.sh predict_ observable	{"stat_name": "CPP", "omega_m": 0.1375, "sigma_ 8": 0.8, "kappa_w": 3, "e_ w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "z": 0, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-...	{ "status": "success", "message": "Predicted CPP at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128/CPP_ z0.0...

#	Tool	Arguments	Outcome
63	serve-hydro.sh predict_observable	{"stat_name": "CPP", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "z": 0.4987, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-clu...	{ "status": "success", "message": "Predicted CPP at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CPP_z0.4...
64	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.0000_prediction.csv"}	# label: CGD z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CGD # redshift: 0.0 # x_label: r/R_{500c} # y_label: $\rho_{\ma...
65	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.4987_prediction.csv"}	# label: CGD z=0.4987 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CGD # redshift: 0.4987 # x_label: r/R_{500c} # y_label: $\rho_{...
66	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.0000_prediction.csv"}	# label: CTP z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CTP # redshift: 0.0 # x_label: r/R_{500c} # y_label: $T / T_{\m...
67	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.4987_prediction.csv"}	# label: CTP z=0.4987 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CTP # redshift: 0.4987 # x_label: r/R_{500c} # y_label: $T / T_...

#	Tool	Arguments	Outcome
68	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/1404.4636.txt", "length": 8000, "start": 24000}	[chars 24000-32000 of 45574] (more available, next start=32000) divided into three equal sized mass accretion rate bins as denoted in the legend, shown in red, green and blue from most slowly to most...
69	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CPP_z0.0000_prediction.csv"}	# label: CPP z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CPP # redshift: 0.0 # x_label: r/R_{500c} # y_label: $P / P_{\m...
70	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CPP_z0.4987_prediction.csv"}	# label: CPP z=0.4987 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) # observable: CPP # redshift: 0.4987 # x_label: r/R_{500c} # y_label: $P / P_...
71	serve-hydro.sh predict_observable	{"stat_name": "CGD", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.02, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-...	{ "status": "success", "message": "Predicted CGD at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...
72	serve-hydro.sh predict_observable	{"stat_name": "CGD", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.02, "z": 0.4987, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-clu...	{ "status": "success", "message": "Predicted CGD at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...

#	Tool	Arguments	Outcome
73	serve-hydro.sh predict_observable	{"stat_name": "CGD", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 1.2, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-p...	{ "status": "success", "message": "Predicted CGD at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...
74	serve-hydro.sh predict_observable	{"stat_name": "CGD", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 1.2, "z": 0.4987, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-clus...	{ "status": "success", "message": "Predicted CGD at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...
75	serve-hydro.sh predict_observable	{"stat_name": "CTP", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.02, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-...	{ "status": "success", "message": "Predicted CTP at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...
76	serve-hydro.sh predict_observable	{"stat_name": "CTP", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.02, "z": 0.4987, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-clu...	{ "status": "success", "message": "Predicted CTP at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...
77	serve-hydro.sh predict_observable	{"stat_name": "CTP", "omega_m": 0.1375, "sigma_8": 0.8, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 1.2, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-p...	{ "status": "success", "message": "Predicted CTP at z=0.0000 (19 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epsk...

#	Tool	Arguments	Outcome
78	serve-hydro.sh predict_ observable	{"stat_name": "CTP", "omega_m": 0.1375, "sigma_ 8": 0.8, "kappa_w": 3, "e_ w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 1.2, "z": 0.4987, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ramp-combo-clus...	{ "status": "success", "message": "Predicted CTP at z=0.4987 (19 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128/d03_ epsk...
79	local cite	{"arxiv_id": "1404.4636", "note": "[\"where the best ?t parameters are A = 0.452± 0.001, B =\\n0.841± 0.008, andγ = 1.628± 0.019\ ", \"This formula provides\ \na good description of the mean pro?le with the accuracy of\\na...	Cited: Hydrodynamic Simulation of Non-thermal Pressure Profiles of Galaxy Clusters (\ cite{arxiv14044636})
80	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128/ 2409.07849.txt", "start": 30000, "length": 15000}	[chars 30000-45000 of 95796] (more available, next start=45000) dian bias above zero, when ex- pressed as a function of?HSE, 500c. The cubic-spline interpolation method is especially susceptible to th...
81	serve-hydro.sh plot_observable_ comparison	{"prediction_files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128/CGD_ z0.0000_prediction.csv", "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-pro...	{ "status": "success", "message": "Plotted 5 CGD predictions.", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-cluster-profiles/ runs/261020_143128/d03_ plots_cgd/CGD_... [Fig. 21]

#	Tool	Arguments	Outcome
82	serve-hydro.sh plot_observable_comparison	{"prediction_files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.0000_prediction.csv", "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-pro...	{ "status": "success", "message": "Plotted 5 CTP predictions.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_plots_ctp/CTP_... [Fig. 22]
83	board post_finding	{"claim": "Using eps_kin (kinetic AGN feedback efficiency, range 0.02-1.2) as the feedback lever, with the other parameters at the design midpoint (kappa_w=3, e_w=0.6, M_seed=1.3, v_kin=0.65, omega_m=0.1375, sigma_8=0.8)...	Recorded finding f-008.
84	board post_question	{"text": "From the d03_epskin_lo/hi CGD and CTP CSVs (z=0 and z=0.4987), what are s_n+s_T and T/T500 at R500c? Is the change in M_HSE/M500 between eps_kin=0.02 and 1.2 smaller or larger than the d-01 redshift change (0.9...	Recorded question q-08.
85	board post_finding	{"claim": "Nelson, Lau & Nagai 2014 (arXiv:1404.4636, Eq. 7, cited verbatim in references.bib) fit P_rand/P_tot(r) = 1 - A{1+exp[-((r/R200m)/B)^gamma]} with A=0.452+-0.001, B=0.841+-0.008, gamma=1.628+-0.019, valid to ab...	Recorded finding f-009.
86	board post_finding	{"claim": "Applying the Nelson et al. f_nth at R500c (0.25-0.29) as a multiplicative factor 1/(1-f_nth) = 1.34-1.42 to the emulator values M_HSE/M500 = 0.925 (z=0) and 0.840 (z=0.4987) from f-005 gives 1.24-1.31 at z=0 a...	Recorded finding f-010.

#	Tool	Arguments	Outcome
87	board post_finding	{"claim": "FLAMINGO (arXiv: 2409.07849) full text, Sec. 3.2-3.4 and Figs. 1, 2, 3 and 5:\n- For 1e13.75 < M500c < 1e14.25 Msun, all five profile methods converge to a median b_ HSE of about -0.1.\n- The FLAMINGO calibratio...	Recorded finding f-011.
88	board post_dead_end	{"what": "Steps 3 (search for documentation of HACC profile mass selection and T weighting) and 4 (X-COP 1805.00042 slope comparison) of d-06.", "why": "Not attempted: the tool budget ran out after reading Nelson and FLA...	Recorded dead end x-02.
89	board post_finding	{"claim": "I checked the pressure slope from CPP (P/ P500, midpoint parameters) against s_n+s_T from CGD and CTP, using centred finite differences with the same bin pairs and independent GP errors. At R500 they agree with...	Recorded finding f-012.
90	board post_finding	{"claim": "I propagated the T(R500)/T500 GP error into M_HSE/M500 = -(T/T500) s/2.\ n\n**Inputs:**\n- T(R500)/ T500 is the geometric mean of the 0.866 and 1.155 R500 bins: 0.5916+-0.0047 at z= 0 and 0.5181+-0.0101 at z= 0.49...	Recorded finding f-013.
91	board post_question	{"text": "Why does CPP (P/ P500) have a much shallower slope than CGD x CTP beyond R500? At 2.05 R500, s_ P = -2.88 against s_n+s_ T = -4.25 at z=0, a 14 sigma difference. Is CPP a per-halo average of pressure (mean of prod...	Recorded question q-09.

A.5 Files produced

- CGD_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.0000_prediction.csv
- 2409.07849.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/
projects/ramp-combo-cluster-profiles/runs/261020_143128/2409.07849.txt
- 1805.00034.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/
projects/ramp-combo-cluster-profiles/runs/261020_143128/1805.00034.txt
- 1404.4636.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/
projects/ramp-combo-cluster-profiles/runs/261020_143128/1404.4636.txt
- CGD_z0.4987_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/CGD_z0.4987_prediction.csv
- CTP_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.0000_prediction.csv
- CTP_z0.4987_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/CTP_z0.4987_prediction.csv
- 2409.05815.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/
projects/ramp-combo-cluster-profiles/runs/261020_143128/2409.05815.txt
- CPP_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/CPP_z0.0000_prediction.csv
- CPP_z0.4987_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/CPP_z0.4987_prediction.csv
- CGD_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_lo/CGD_z0.0000_
prediction.csv
- CGD_z0.4987_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_lo/CGD_z0.4987_
prediction.csv
- CGD_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_hi/CGD_z0.0000_
prediction.csv
- CGD_z0.4987_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_hi/CGD_z0.4987_
prediction.csv
- CTP_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_lo/CTP_z0.0000_
prediction.csv
- CTP_z0.4987_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing
projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_lo/CTP_z0.4987_
prediction.csv

- `CTP_z0.0000_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testing projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_hi/CTP_z0.0000_ prediction.csv`
- `CTP_z0.4987_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testing projects/ramp-combo-cluster-profiles/runs/261020_143128/d03_epskin_hi/CTP_z0.4987_ prediction.csv`

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
`d-01`	closed	Use the HACC hydro emulator to predict CGD (gas density) and CTP (temperature) profiles at two redshifts for a fiducial parameter set. Derive the log-slopes d ln n_gas/d ln r and d ln T/d ln r at 0.5, 1 and 2 R500c, and from them a hydrostatic mass estimate and b_HSE at each redshift.	This direction is done except for one item: the separate plot_prediction figure with an uncertainty band for each observable was not made, because the tool budget ran out. The redshift pair is z=0 vs z=0.4987, not z=0 vs z=1, because the CGD and CTP profiles only go up to z=0.4987. The emulator has no fiducial parameter set, so I used the midpoint of every design range: kappa_w=3, e_w=0.6, M_seed=
`d-02`	closed	Build a cited literature baseline on hydrostatic mass bias and its redshift and feedback dependence, with quotable numbers to compare against the emulator results.	Partly done. The literature baseline is limited to what the paper abstracts say. Four papers were added to references.bib with verbatim quotes checked by cite (tc-...-48 to -51); the success criterion was at least six, so this falls short. The file is /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/references.bib. Established values, posted as findi

Id	Status	Goal	Outcome
d-03	blocked	Test how robust the profile-shape-driven bias evolution is to the emulator's astrophysical (feedback) and cosmological parameters. Estimate the non-thermal pressure fraction needed to reconcile M_HSE with the true mass.	Partly done. The predictions and comparison plots for the feedback test exist, but I did not read the numbers out of them, so none of the slope or mass results asked for are available yet. The feedback lever I chose is eps_kin (kinetic AGN feedback efficiency), whose allowed range is 0.02 to 1.2 (tc-...-61). This is a judgement call: the emulator does not say which parameter controls the density s
d-04	closed	Test the mass dependence of the profile-slope-driven HSE bias, groups versus clusters, at both redshifts. This lets the FLAMINGO result (2409.07849) of groups rising and clusters falling with z be compared directly.	I could not test mass dependence because the emulator has no mass input. list_observables (tc-261020_1-w1-d-04-20) and describe_parameters (tc-261020_1-w1-d-04-21) show that CGD, CTP and CPP take only redshift and 7 parameters: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m and sigma_8. One test prediction (CGD at z=0 with fiducial mid-range parameters; file /Users/nramachandra/hep-genesis-test
d-05	closed	Test whether the change in M_HSE/M500 between z=0 and z=0.4987 at the midpoint parameters is statistically robust. Cross-check s_n + s_T against the emulator's own pressure profile (CPP), propagate the T(R500)/T500 error, and restate the result in the FLAMINGO sign convention.	The redshift change in M_HSE/M500 is not statistically robust: it is about 1.2 sigma whichever way the slope is computed. I predicted CPP (P/P500) at the d-01 midpoint parameters (tc-...-62, -63). The files are CPP_z0.0000_prediction.csv and CPP_z0.4987_prediction.csv in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-cluster-profiles/runs/261020_143128/. The CGD, CTP and CPP bin value

Id	Status	Goal	Outcome
d-06	closed	Extract quantitative literature benchmarks from the full texts already downloaded. Use them to compare the z=0 emulator slopes with X-COP and FLAMINGO, and to compute the Nelson et al. non-thermal correction at R500c. Also search once for documentation of the HACC profile mass selection and temperature weighting.	Partly done: the Nelson et al. benchmark is extracted, cited and applied, and FLAMINGO results were read from the full text. Three things remain open. (a) The FLAMINGO quotes were not added to references.bib with cite. (b) The search for HACC mass-selection and temperature-weighting documentation was not run, so q-01, q-03 and q-06 remain open. (c) The X-COP 1805.00042 slope comparison was not don

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	high	FLAMINGO (Braspenning et al. 2024, arXiv:2409.07849): fiducial median b_HSE = (M_HSE,500c - M500c)/M500c goes from about -0.1 (groups) to about -0.2 (clusters). Scatter at fixed true mass is smallest, sigma(b) about 0.1, at M500c ~ 1e14 Msun. At fixed true mass, HSE masses rise with z for groups and fall with z for clusters, and the scatter grows. Models with lower gas fractions give more-biased group masses and less-biased cluster masses. The bias does not depend on the fitting function, but it does depend on fit quality: poorer fits give a stronger bias. The scatter comes from differences in	tc-261020_1-w1-d-02-40
f-002	high	X-COP (Eckert et al. 2018, arXiv:1805.00034), 13 nearby clusters with XMM+Planck: median non-thermal pressure fraction ~6% at R500 and ~10% at R200. This is lower than hydrodynamical simulations predict. With the Planck mass calibration, the gas fractions imply 1-b = 0.85 +/- 0.05 for SZ masses, with some evidence for a mass-dependent bias.	tc-261020_1-w1-d-02-41

Id	Confidence	Claim	Evidence (tool calls, figures)
f-003	high	Nelson, Lau & Nagai 2014 (arXiv:1404.4636): the non-thermal pressure fraction profile is universal across redshift when radii are scaled to the mean matter density (R200m), not to the critical density. It depends strongly on mass accretion rate, and the non-thermal pressure is mainly radial. A fitting formula for f_nth(r/R200m) is given; parameter values are not yet extracted.	tc-261020_1-w1-d-02-42
f-004	high	Simba (Sorini et al. 2024, arXiv:2409.05815), six feedback variants, 1e11 < M200c < 1e14 Msun, 0 < z < 4: gas-density profiles follow a power law. At fixed halo mass, slope and amplitude depend only weakly on redshift. With AGN jets on, haloes above M200c ~ 1e13 Msun have shallower (gentler) density profiles. A universal fitting formula for slope(M, z) and amplitude(M, z) is provided.	tc-261020_1-w1-d-02-43
f-005	medium, **contested**	HACC emulator at design-midpoint parameters (kappa_w=3, e_w=0.6, M_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8). Profiles are only available for z<=0.4987, so the pair used is z=0 and z=0.4987. CGD is rho_gas/rho_crit(z) and CTP is T/T500, on 19 log bins (0.015-2.74 R500c). Centred log-slopes, with 1-sigma errors from the GP std assuming independent bins. At z=0: s_n = -1.86+-0.10, -2.55+-0.16, -3.13+-0.07 and s_T = -0.27+-0.03, -0.58+-0.06, -1.12+-0.05 at 0.49, 1.0 and 2.05 R500c. At z=0.4987: s_n = -1.87+-0.10, -2.74+-0.10, -2.93+-0.06 and s_T = -0.35+-0.06, -0.50+-0.14,	tc-261020_1-w1-d-01-22, tc-261020_1-w1-d-01-35, tc-261020_1-w1-d-01-37, tc-261020_1-w1-d-01-38, tc-261020_1-w1-d-01-39
f-006	medium	Barranco-Llorca et al. 2025 (arXiv:2511.10730), AMR cluster mergers at 0 <= z <= 1.5: at z=0 the HSE bias correlates only mildly with dynamical state. During a major merger the bias dips sharply negative around the merger time, then shows a transient positive peak, then slowly returns to its pre-merger level. This is driven mainly by changes in the gas-density morphology and dynamics, not by thermodynamics, and it shows weak dependence on mass ratio and impact parameter.	tc-261020_1-w1-d-02-54

Id	Confidence	Claim	Evidence (tool calls, figures)
f-007	high	Comparison overlays of the HACC emulator profiles at z=0 and z=0.4987 (midpoint parameters). In CGD (rho_gas/rho_crit(z)) the two redshifts agree within about 3% over 0.2-0.9 R500c. At z=0.4987 the density is about 6% lower at 1.15 R500c and about 13-14% lower at 1.5-2.05 R500c, then converges again at 2.74 R500c, and the core (r<0.1 R500c) is about 15-50% lower. In CTP (T/T500), z=0.4987 sits 7-13% below z=0 from 0.15 R500c to 1.15 R500c (12% lower at R500c itself), and the gap widens to 30% by 2.74 R500c.	tc-261020_1-w1-d-01-53, tc-261020_1-w1-d-01-55, tc-261020_1-w1-d-01-35, tc-261020_1-w1-d-01-37, tc-261020_1-w1-d-01-38, ... Fig. 9, 10
f-008	high	Using eps_kin (kinetic AGN feedback efficiency, range 0.02-1.2) as the feedback lever, with the other parameters at the design midpoint (kappa_w=3, e_w=0.6, M_seed=1.3, v_kin=0.65, omega_m=0.1375, sigma_8=0.8), the HACC emulator produced CGD and CTP profiles at z=0 and z=0.4987 for eps_kin=0.02 and 1.2. They are plotted against the d-01 midpoint (eps_kin=0.61, z=0) in Figure 21 (CGD) and Figure 22 (CTP). The numerical slopes s_n+s_T at R500c have NOT yet been extracted from these files.	tc-261020_1-w2-d-03-61, tc-261020_1-w2-d-03-71, tc-261020_1-w2-d-03-72, tc-261020_1-w2-d-03-73, tc-261020_1-w2-d-03-74, ... Fig. 21, 22
f-009	medium, **contested**	Nelson, Lau & Nagai 2014 (arXiv:1404.4636, Eq. 7, cited verbatim in references.bib) fit P_rand/P_tot(r) = 1 - A{1+exp[-((r/R200m)/B)^gamma]} with A=0.452+-0.001, B=0.841+-0.008, gamma=1.628+-0.019, valid to about 10% over 0.1-1.5 R200m out to z=1.5 (non-radiative ART runs, mass-weighted T and sigma). Evaluated at R500c: x = r/R200m = 0.5, 0.55 and 0.6 gives f_nth = 0.254, 0.274 and 0.294, with 1/(1-f_nth) = 1.34, 1.38 and 1.42. Assumption, computed by hand not with a tool: R500c is about 0.5-0.6 R200m, nearer 0.5 at z=0 and nearer 0.55-0.6 at z=0.5 because Omega_m(z) is higher. Because the pro	tc-261020_1-w2-d-06-68, tc-261020_1-w2-d-06-79, tc-261020_1-w2-d-06-60

Id	Confidence	Claim	Evidence (tool calls, figures)
f-010	low, **contested**	Applying the Nelson et al. f_nth at R500c (0.25-0.29) as a multiplicative factor 1/(1-f_nth) = 1.34-1.42 to the emulator values M_HSE/M500 = 0.925 (z=0) and 0.840 (z=0.4987) from f-005 gives 1.24-1.31 at z=0 and 1.16-1.19 at z=0.4987. That is a 16-31% overcorrection. To reconcile the emulator directly, f_nth would need to be only about 0.075 (z=0) and 0.16 (z=0.4987). These values are closer to the X-COP estimate (about 6% at R500, f-002) than to the non-radiative Nelson fit. So either the emulator's T500 convention or its temperature weighting already absorbs part of the bias, or the HACC pro	tc-261020_1-w2-d-06-68, tc-261020_1-w1-d-01-38, tc-261020_1-w1-d-01-39
f-011	medium, **contested**	FLAMINGO (arXiv:2409.07849) full text, Sec. 3.2-3.4 and Figs. 1, 2, 3 and 5: - For 1e13.75 < M500c < 1e14.25 Msun, all five profile methods converge to a median b_HSE of about -0.1. - The FLAMINGO calibration assumed a constant b of -0.257, taken from weak lensing, which agrees with the predictions for the most massive clusters. - Fig. 2: volume-weighted profiles give a significantly smaller and less mass-dependent bias than X-ray-weighted ones, and mass-weighted profiles fall in between. The authors conclude that for M500c >> 1e13 Msun most of the bias comes from X-rays being a biased tracer	tc-261020_1-w2-d-06-80
f-012	high	I checked the pressure slope from CPP (P/P500, midpoint parameters) against s_n+s_T from CGD and CTP, using centred finite differences with the same bin pairs and independent GP errors. At R500 they agree within 0.4 sigma. At 0.49 R500 they differ by 1.3 sigma (z=0) and 2.4 sigma (z=0.4987). At 2.05 R500 they disagree strongly, by 14 sigma (z=0) and 11 sigma (z=0.4987). | Radius | z | s_P (CPP) | s_n+s_T | Difference | |—|—|—|—|—| | 0.49 R500 | 0 | -1.981+-0.042 | -2.133+-0.104 | 1.3 sigma | | 0.49 R500 | 0.4987 | -1.874+-0.088 | -2.218+-0.117 | 2.4 sigma | | 1.00 R500 | 0 | -3.052+	tc-261020_1-w2-d-05-62, tc-261020_1-w2-d-05-63, tc-261020_1-w2-d-05-64, tc-261020_1-w2-d-05-65, tc-261020_1-w2-d-05-66, ...

Id	Confidence	Claim	Evidence (tool calls, figures)
f-013	medium	I propagated the T(R500)/T500 GP error into M_HSE/M500 = -(T/T500) s/2. **Inputs:** - T(R500)/T500 is the geometric mean of the 0.866 and 1.155 R500 bins: 0.5916+-0.0047 at z=0 and 0.5181+-0.0101 at z=0.4987. **Results with s = s_n+s_T:** - z=0: 0.925+-0.051. - z=0.4987: 0.840+-0.048. - Difference: 0.085+-0.070, which is 1.2 sigma. **Results with s = s_P from CPP (independent second estimate):** - z=0: 0.903+-0.034. - z=0.4987: 0.837+-0.043. - Difference: 0.066+-0.055, which is 1.2 sigma. **In the FLAMINGO convention, b = (M_HSE-M500)/M500:** - n+T route: b = -0.075+-0.051 (z=0) and -0.160	tc-261020_1-w2-d-05-64, tc-261020_1-w2-d-05-65, tc-261020_1-w2-d-05-66, tc-261020_1-w2-d-05-67, tc-261020_1-w2-d-05-69, ...

B.3 Dead ends

- x-01 (d-04): Predicting CGD/CTP at group-scale (~1e13.5 Msun) and cluster-scale (~1e14.5-15 Msun) masses to test the mass dependence of the slope-driven HSE bias. — The CGD/CTP/CPP emulators have no mass input. The only inputs are the 7 parameters kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m and sigma_8, plus z. The output is one 19-bin profile in r/R500c (0.0154-2.74), with y = rho_gas/rho_crit and no mass column or mass bins, so each profile is for a single implicit (undocumented) cluster mass selection. The cluster profiles are also trained only for z
- x-02 (d-06): Steps 3 (search for documentation of HACC profile mass selection and T weighting) and 4 (X-COP 1805.00042 slope comparison) of d-06. — Not attempted: the tool budget ran out after reading Nelson and FLAMINGO. The steps are not impossible; q-01, q-03 and q-06 and the X-COP slope benchmark stay open.

B.4 Critic verdicts

- c-01 (wave 0, plan): **revise**; high: The calls do not fit the 13-call budget. Planned calls are list_observables, describe_parameters, 4 predict_observable, list_agent_files, up to 4 load_table_data, preview, describe, 2 overlays and 2 plot_prediction, which is about 18. Cut preview and describe_dataset, use one plot_prediction or none; high: b_HSE needs a true mass, and that has to come from the emulator definition, not from assumption. Check whether CGD and CTP are stacked in M500c bins and whether radii are already scaled by the true R500c. If so, b_HSE at r = R500c(true) is well defined. If not, report only M_HSE(z_high)/M_HSE(z_low); medium: Check that 0.5, 1 and 2 R500c lie inside the emulator's radial range and grid. If 2 R500c is outside, say so rather than extrapolating. Differencing the upper and lower band edges does not give a valid slope uncertainty, because band edges are not correlated realisations. Label it a crude bound, or ; medium: list_observables reports a CPP (pressure) profile. Use d ln P/d ln r from CPP as an independent cross-check of s_n + s_T at both redshifts (2 extra predict calls; budget allowing). Without it the HSE slope rests on two separately

emulated profiles whose errors may not be consistent.; medium: Six fully cited papers with download, read and cite each takes about 18-20 calls, which is over budget. Fully cite 4 core papers: 2409.07849, 1805.00034 or 1805.00035, 1404.4636 and 2409.05815. Cover the rest with search_arxiv_abstracts and get_arxiv_metadata as abstract-level summaries, and do not ; low: Note that the emulator is HACC-based, not CAMELS (2010.00619), and FLAMINGO, Simba and HACC differ in subgrid models. Comparisons should be framed as qualitative trends, not one-to-one numbers.

- `c-02` (wave 1, wave): **continue**; contested `f-005`; high: The redshift change in M_HSE/M500 (0.925 vs 0.840) is about 1.3 sigma once errors are combined, and T(R500)/T500 errors are not propagated. The summary's statement that the larger bias at z=0.4987 comes from the 12% lower T(R500) needs a significance test before it enters the final discussion. Re-qu; medium: Sign convention mismatch. f-005 reports b_HSE as positive (1 - M_HSE/M), while FLAMINGO (f-001) uses (M_HSE - M)/M, which is negative. Harmonise the convention before comparing the two. Also note that if the profile stack uses a fixed mass threshold, the mean mass is lower at z=0.4987, so T/T500 and; medium: The critique c-01 suggestion to cross-check s_n + s_T against d ln P/d ln r from CPP was not done. Neither was a smooth-fit derivative (q-04). Finite differences of GP means with assumed independent errors are a weak basis for the 3.6-sigma s_T claim at 2 R500c.; low: The core difference (15-50% lower at z=0.4987) sits inside very large emulator errors. For example, at the innermost bin z=0.4987 gives 1018 +- 689 against 2154 +- 258 at z=0. It should be labelled not significant. The outer-radius percentages are not visible in the quoted excerpts, though the figur; low: f-001 to f-004 cite truncated read_paper excerpts that show only title and author headers. The verified cite calls (tc-48 to tc-51) should be attached as evidence so the quoted numbers can be checked directly. f-006 is abstract-only and uncited.

- `c-03` (wave 2, wave): **continue**; contested `f-009`, `f-010`, `f-011`; high: The feedback robustness test, the main open robustness question, has produced files but no numbers. Until the slopes and M_HSE/M500 are extracted for eps_kin=0.02 and 1.2, there is no evidence on whether Delta b(z) is larger or smaller than the feedback-induced spread. f-008 is supported only as a r; medium: The quoted CSV excerpts end at the first ~7 bins ($r < 0.09$ R500c), before the 0.365-2.738 R500c bins used for the slopes. The values are internally consistent: s_n+s_T reproduces d-01 exactly, and the errors add in quadrature correctly. Even so, the specific bin values should be quoted. The 14-sigma; medium: The 1.2-sigma significance treats GP errors as independent across bins and across redshifts. The z=0 and z=0.4987 predictions come from the same emulator, so correlated errors could raise or lower the significance substantially. Also, T(R500)/T500 = 0.59 is low compared with observed universal profi; low: The FLAMINGO quotes still need to be recorded with cite. The X-COP 1805.00042 slope benchmark and the HACC documentation search (q-01, q-03, q-06) were not run.

- `c-04` (wave 2, report): **ok**; high: unsupported: The density slope does not evolve significantly at any of the three radii, in line with the Simba expectation. — The board does not make this claim. At 2.05 R500c the board's own values (-3.13+-0.07 vs -2.93+-0.06) differ by about 2.2 sigma under the stated independent-error treatment, ; high: unsupported: 4. Refit the profiles with smooth analytic models and propagate correlated emulator errors to re-assess the significance of the temperature-slope steep — The report is truncated mid-sentence, so the final open direction and any closing material are incomplete. This is not an unsupported